(a) $\log_{10}$ peak $\Omega_{\rm GW}h^2$, clusters with burst efficiency

(b) $\log_{10}$ (margin below $\Delta N_{\rm eff}$), NL-B with burst efficiency

(c) $\log_{10}$ peak $\Omega_{\rm GW}h^2$, shot noise, Choptuik

(d) $\log_{10}$ $\max_{\rm det}\Omega/\Omega_{\rm PLI}$ (SNR 1, 1 yr), clusters

red dashed: published induced-GW bound
black dotted: our linear extrapolation
reaches $\Delta N_{\rm eff}$ (damping on)
grey: β_c (PBH domination)
hatched: merged relics (Holst et al.)
blue dash-dot: half the holes merge (estimate)
blue dotted: a quarter merge; merged holes re-dominate

DECIGO, SNR 10
BBO, SNR 10
ET, SNR 10
CE, SNR 10
DECIGO, shock-limited
BBO, shock-limited

$\log_{10}\beta_f$

$\log_{10} M_{\rm in}$ [g]